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MASTER THESIS REPORT

Is the Best, the Best? Analysing Best-Paper Awardees' Career Trajectories

Student:

Shaheryar Shaheryar
shaheryar.4822@student.uu.se

Supervisors:

Georgios Panayiotou
georgios.panayiotou@it.uu.se

Subject Reviewer:

Arjun Menon
arjun.menon@it.uu.se

Department of Information Technology

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1 Introduction

In almost every computer science conference, a small number of papers are singled out as “best.” Committees deliberate, a few teams walk onto a stage, and CVs gain a new line. What happens after the applause, however, is much less clear. Do such awards change the careers of the junior researchers who receive them, or do they mostly confirm success that was already on the way?

Recognition and rewards play a central role in how modern science is organised. Formal evaluation systems like tenure decisions, grant competitions, hiring processes rely heavily on signals of excellence such as publications, citations, and prizes. Informally, too, awards shape how researchers are perceived. They can open doors to collaborations, make job applications stand out, or simply change how colleagues read a paper. Yet the relationship between recognition and subsequent achievement is not straightforward. Studies of major prizes such as the Nobel Prize or the Fields Medal show that awards can be followed by shifts in focus and lower citation impact per paper for senior scientists, suggesting that being celebrated may change how they work and what they work on (1; 2). Other work, focusing on earlier recognition, finds that prize winners subsequently publish more, receive more citations, and remain active longer than closely matched peers who did not receive awards. Together, these findings suggest that awards may function both as markers of past excellence and as interventions that reshape scientific trajectories.

This raises broader questions about fairness and opportunity in academia. If early recognition helps some researchers accelerate their careers, while others with similar potential never get that push, awards may amplify existing inequalities rather than merely recording achievement. If, on the other hand, awards mostly identify people who were already on fast trajectories, their practical impact on careers may be limited. Understanding which of these possibilities is closer to reality matters not only for those who win or miss out, but also for universities, funders, and conference organisers who increasingly rely on awards in their evaluation practices. Most existing research on these issues focuses on rare, high-profile prizes aimed at established scientists. In computer science, however, much of the evaluation of research happens at the conference level. Conferences¹ are primary publication venues in many subfields, and best paper awards are granted far more frequently than global prizes. They are often given to teams that include doctoral students or recent graduates, at a moment when careers are still taking shape. For a junior researcher, a best paper award can be the first visible sign that their work is making a mark. Despite this, little is known about whether such conference-level awards are associated with distinctive career trajectories. Do junior awardees go on to publish more, receive more citations, or produce more disruptive work than similar colleagues whose papers did not receive awards? Or do their careers look much the same once we account for where they publish and how strong their early record already was?

¹<https://www.acm.org/conferences>

This thesis addresses that gap by analysing the career trajectories of junior authors who receive best paper awards at computer science conferences and comparing them with matched non-awarded peers from the same venues and years. The focus is on early career, defined as the first few years after a researcher’s initial indexed publication, when opportunities and constraints may have long-lasting consequences. By “career trajectory” we mean patterns in measurable outputs over time. How many papers someone publishes per year, how often these papers are cited, how disruptive their contributions are relative to existing work, and how their collaboration networks evolve. These are necessarily imperfect proxies for concepts such as quality or creativity, but they capture aspects of visibility and uptake that are central to how academic careers are evaluated in practice.

The project builds on several strands of prior research. Studies of major prizes suggest that recognition can be followed by both positive and negative shifts in productivity and impact, depending on the timing and nature of the award. Work on early-career recognition shows that scientists who receive competitive prizes relatively early tend to publish more, receive more citations, and remain active longer than comparable non-winners, even when their initial performance was similar (3). Other research highlights how a small number of early-career features such as early citations, institutional prestige, and collaborations with established researchers strongly predict long-term impact across fields (4). At the same time, studies of team size and collaboration patterns show that small, focused teams are more likely to produce disruptive work that changes how later papers cite the literature, while larger teams tend to develop and extend established lines of research (5). These results suggest that any effect of conference awards on junior researchers is likely to be intertwined with where they publish, who they work with, and how their work is used by others.

Against this backdrop, the thesis asks three main questions. First, how do the publication and citation trajectories of junior best-paper awardees differ from those of similar non-awardees from the same conferences and years? Second, to what extent do awardees and non-awardees diverge in the disruptiveness of their subsequent work, as captured by citation-based indices that distinguish between contributions which redirect attention away from prior work and those which consolidate existing streams? Third, are any observed differences concentrated in particular venues, subfields, or moments in a career, or are they relatively consistent across contexts? An optional fourth line of inquiry concerns co-authorship networks. Whether the structure and diversity of collaborations for awardees evolves differently over time than for matched peers.

By combining large-scale bibliometric data from OpenAlex² with curated lists of best paper awards, and by comparing awardees with carefully matched non-awardees, the thesis aims to provide an empirical account of how conference-level recognition relates to junior researchers’ subsequent paths in computer science. In doing so, it contributes both to the science-of-science literature on rewards and careers and to practical debates about

²<https://docs.openalex.org>

how we design and interpret recognition systems in academia.

2 Methods

This study uses quantitative, observational methods grounded in scientometrics and network science to compare the careers of junior best-paper awardees with those of similar non-awardees. The empirical context is the computer science research community, where conferences are primary publication venues and career progression is often evaluated through bibliometric indicators rather than direct observation of workplace practices. Data are therefore collected from large-scale scholarly databases and curated award lists rather than from labs, workplaces, or classrooms.

The core data source is OpenAlex, an open index of scholarly works, authors, venues, institutions, and citation relations. For a selected set of computer science conferences and years, the study retrieves metadata on all papers and their authors, including publication year, venue, citation counts, reference lists, and authorship information. These records are linked to curated best paper award lists from ACM and community-maintained collections, which provide conference, year, title, authors, and award type for each recognised paper. Matching award entries to OpenAlex work records yields a cohort of awarded papers and their junior authors.

The main units of analysis are authors who receive at least one best paper award within five years of their first indexed publication (junior awardees) and a matched set of non-awarded junior authors from the same conferences and years. Matching is performed on venue, award year, and early-career stage, and will be refined using simple propensity-score or exact matching based on pre-award productivity and citations. For each author, the study constructs a longitudinal record of outputs and collaborations before and after the award (or corresponding year for controls).

Key variables include annual publication count (productivity), annual citation count and field-normalised citation rates (impact), team size and co-author diversity (collaboration), and a disruption score based on the CD index, which compares how often later papers cite a focal work without versus alongside its referenced prior work (6). These variables are computed for several years before and after the award event, allowing trajectories to be compared between awardees and controls.

Statistical analyses will combine descriptive summaries with inferential tests. Two-sample tests (such as t-tests or Mann–Whitney U tests) will be used to compare median productivity and citation measures between groups, while regression models controlling for pre-award performance will assess whether award status is associated with different post-award trajectories. Where data permit, stratified analyses by venue, subfield, and institutional context will explore heterogeneity in outcomes. All data processing and analysis will

be implemented in Python using open-source libraries, with code and derived datasets documented to support transparency and reproducibility.

Ethical considerations focus on data quality, fairness, and responsible interpretation. Only publicly available bibliometric metadata and award lists are used. No sensitive personal data are collected. Limitations of author disambiguation and coverage are documented, and results are interpreted as patterns in visibility and impact, not as definitive judgments of individual merit.

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